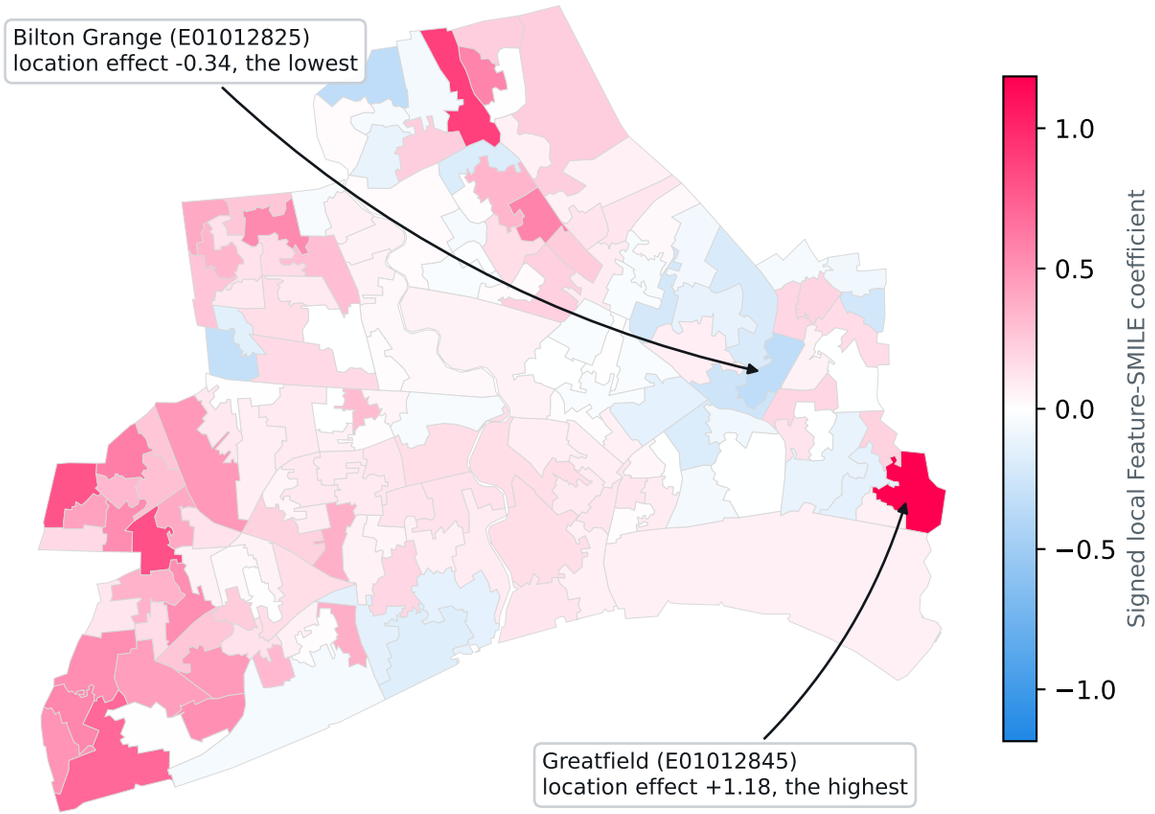
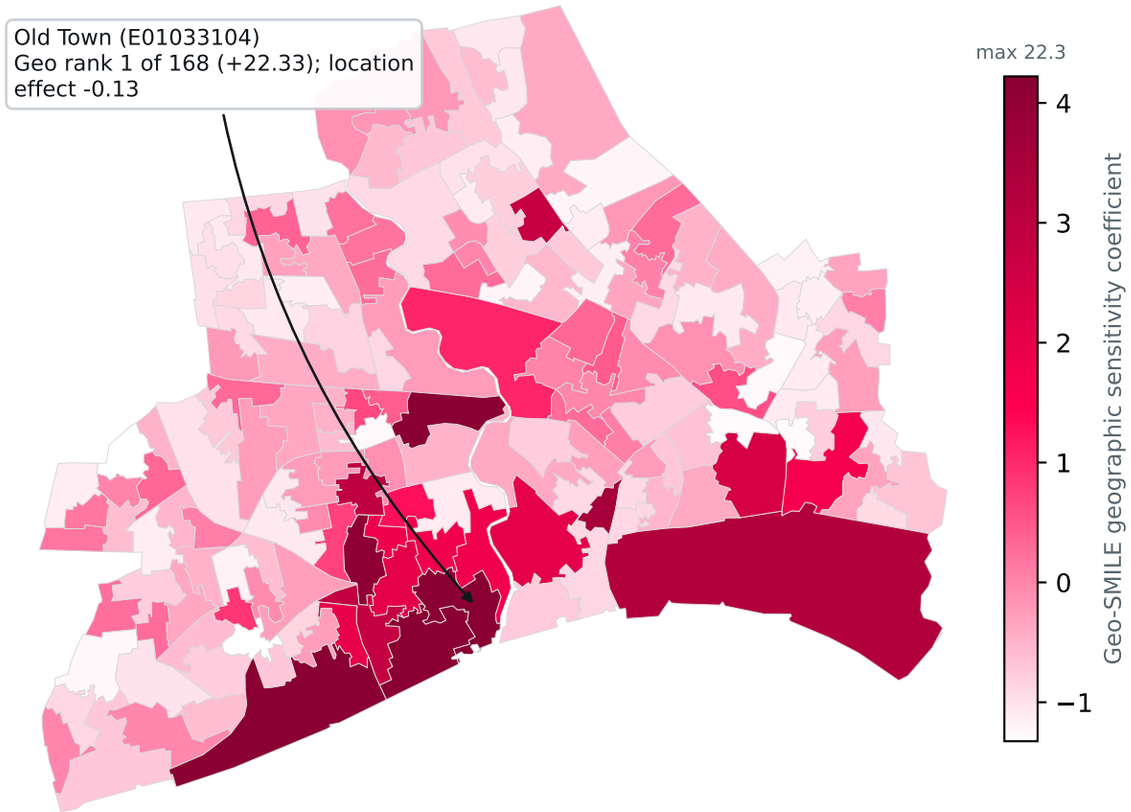
Hull: location as a joint feature player vs the Geo branch

Feature-B: joint location player (signed local attribution)



Geo branch, area-data design (per-LSOA geographic sensitivity)



168 LSOAs, XGBoost, test period. Left: Feature-B SMILE with 31 players (30 predictors + one joint location player masking both centroid coordinates; 4,000 masks, seed 42); local coefficients are fitted to signed prediction differences, so red/blue are opposite directions. Right: area-data Geo-SMILE coefficient (N=4,000), one sequential scale because its sign is not a direction. Spearman -0.17 signed; -0.16 between |location effect| and the Geo coefficient. Not causal.